

environment due to price momentum. Both broker models may be accurate for their particular client base, but if the model is applied to the other broker's clients the model would provide inaccurate estimates.

Investors also need to ensure that the brokers they solicit cost estimates from have properly corrected for multicollinearity and heteroscedasticity. Analysts are encouraged to explore these correction methods with their brokers and vendors.

Table 11.1 shows cost estimates for two stocks from three brokers. Notice how these estimates can vary across brokers. For example, the cost estimates for trading 1% ADV of RLK using a $POV = 20\%$ strategy is 6.0 bp from broker I, 11.4 bp from broker II, and 4.7 bp from broker III. For an order of 30% ADV using a $POV = 5\%$ strategy the estimates are 31.0 bp from broker I, 17.4 bp from broker II, and 21.0 bp from broker III. Broker II had the highest cost estimate in the first situation and the lowest cost estimate in the second situation. It is not uncommon for broker cost estimates to vary to this extent.

Step III: Estimate Model Parameters

The third step of the process is to estimate the model parameters. Since vendors will have positive cost estimates we are able to log-transform our model and solve for the parameters using ordinary least squares (OLS). Since OLS regression analysis is well understood, analysts can easily critique and evaluate the performance of these models.

If any broker or vendor provides a negative cost estimate, meaning you will profit on the trade immediately, it may be time to find a different broker!

The log-transformed simplified I-Star model is:

$$\ln(MI_{bp}) = \ln(a_1) + a_2 \cdot \ln(Size) + a_3 \cdot \ln(\sigma) + a_4 \cdot \ln(POV) \quad (11.2)$$

Estimates for the model parameters obtained using data from Table 11.1 are shown in **Table 11.2**.

The best fit log-transformed model is:

$$\ln(MI_{bp}) = 6.66 + 0.57 \cdot \ln(Size) + 0.78 \cdot \ln(\sigma) + 0.52 \cdot \ln(POV) \quad (11.3)$$

The model has a high R^2 ($R^2 = 0.93$), significant t-stats ($t \gg 0$), and a high F-stat ($F = 513.47$). This indicates that this is a very reasonable model. If the regression results had low R^2 and/or insignificant t-stats or F-stat, then the proposed analytical form of the preferred model (see step I) should not be used. The good thing is that since most pre-trade models include size,